

**Task No.: 8****Date: 08-02-2026**

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**Title:** Implement an application that lists sensors.**Aim:**

To develop an Android Application that implements the list of sensors available.

**Code:***MainActivity.kt*

```
package com.example.madtask9
```

```
import android.content.Context
import android.hardware.Sensor
import android.hardware.SensorManager
import android.os.Bundle
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    lateinit var txtSensors: TextView
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
        txtSensors = findViewById(R.id.txtSensors)
```

```
        val sensorManager =
```

```
            getSystemService(Context.SENSOR_SERVICE) as SensorManager
```

```
        val sensorList: List<Sensor> =
```

```
            sensorManager.getSensorList(Sensor.TYPE_ALL)
```

```
        val stringBuilder = StringBuilder()
```

```

stringBuilder.append("Available Sensors:\n\n")

for (sensor in sensorList) {

    stringBuilder.append(sensor.name).append("\n")

}
val magnetometer =
    sensorManager.getDefaultSensor(Sensor.TYPE_MAGNETIC_FIELD)

if (magnetometer != null) {

    stringBuilder.append("\nMagnetometer is available")

} else {

    stringBuilder.append("\nMagnetometer is NOT available")

}

txtSensors.text = stringBuilder.toString()

}
}

```

*activity\_main.xml*

```

<?xml version="1.0" encoding="utf-8"?>

<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/txtTitle"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name: Yoga Shri P K&#10;Roll No.: 23BCS192"

```

```
android:gravity="center"  
android:textSize="22sp"  
android:textStyle="bold"  
android:padding="16dp"/>
```

```
<Space  
    android:layout_width="match_parent"  
    android:layout_height="10dp"/>
```

```
<ScrollView  
    android:layout_width="match_parent"  
    android:layout_height="match_parent">
```

```
<TextView  
    android:id="@+id/txtSensors"  
    android:layout_width="match_parent"  
    android:layout_height="wrap_content"  
    android:text="Sensor list will appear here"  
    android:textSize="16sp"  
    android:padding="16dp"/>
```

```
</ScrollView>
```

```
</LinearLayout>
```

## Output:

Name: Yoga Shri P K  
Roll No.: 23BCS192

### Available Sensors:

LSM6DSL Accelerometer  
MXQ4300S Magnetometer  
LSM6DSL Gyroscope  
STK3351z Light  
STK3351z Proximity  
MXQ4300S Magnetometer  
Uncalibrated  
LSM6DSL Gyroscope Uncalibrated  
Significant Motion  
Step Detector  
Step Counter  
Tilt Detector  
Pick Up Gesture  
Device Orientation  
Interrupt Gyroscope  
Scontext  
STK3351z Light CCT  
Call Gesture  
Wake Up Motion  
Pocket Mode  
Device Orientation Wake Up  
SBM  
A96T3X6 Grip sensor  
Grip sensor for sub  
Grip Notifier Sensor  
Motion Sensor  
Samsung Game Rotation Vector  
Sensor  
Samsung Gravity Sensor  
Samsung Linear Acceleration Sensor  
Samsung Rotation Vector Sensor  
Samsung Orientation Sensor  
  
Magnetometer is available

## Result:

Thus, Android Application that implements the list of sensors available is developed and executed successfully.